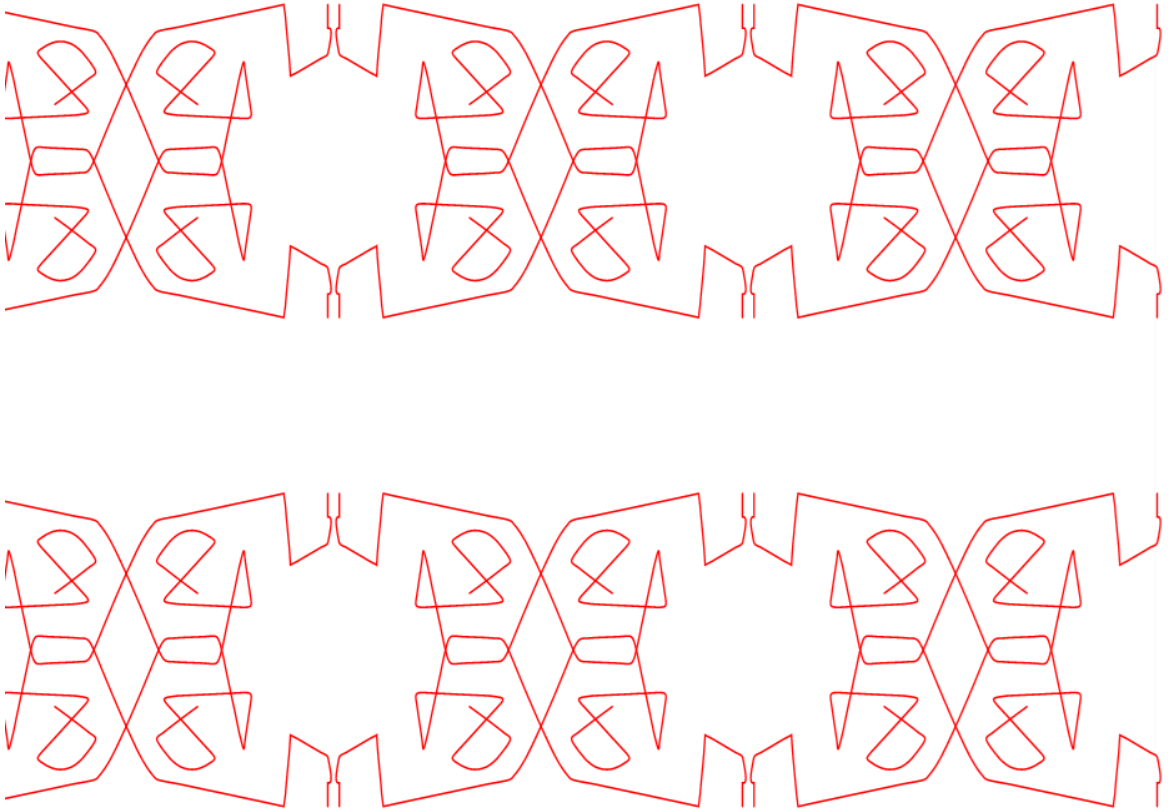
EE2021 -
Materials Science for Electrical Engineers

Crystal Structure

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Is there symmetry in this jumble?



History of Crystals



Copper sulfate crystal - Wikipedia

Gilbert, 1600: “Lucid gems are made of water which is somehow concreted by cold or by frost through the influence of various ill-defined substances”

Hooke 1665: Sphere as a basic unit for natural shapes of crystals

Grignon, 1775: Observed that iron polycrystal is “made of single crystals”

Laue, 1912: X-ray diffraction, experimental proof of crystals being ordered arrangement of atoms

Planar facets of minerals fascinated humans

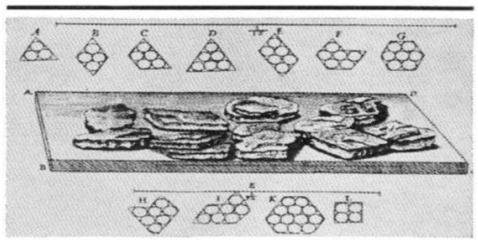
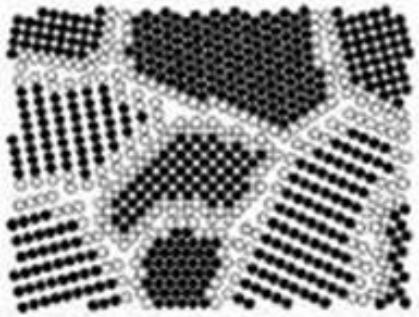


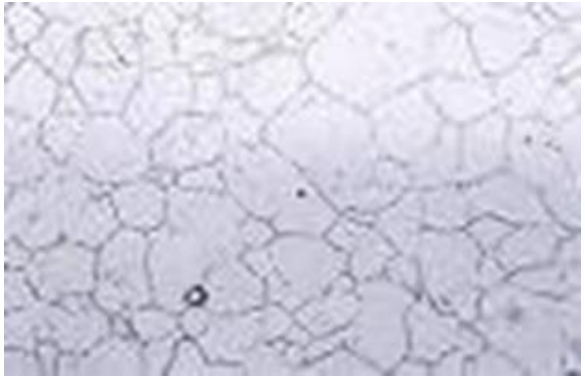
Fig. 3. The sphere-packing arrangements of Hooke (1665) to account for the natural shapes of crystals. (Courtesy of University of California Press).



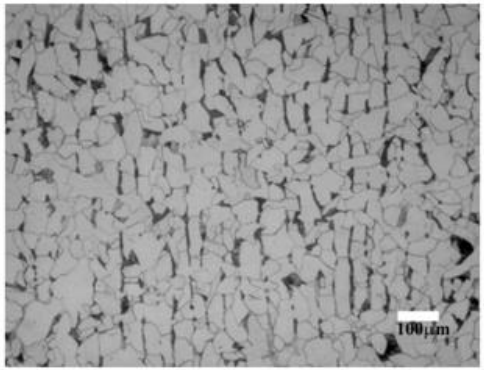
Fig. 1. Drawing by Grignon (1775) of a cluster of iron crystals. (Courtesy of University of Chicago Press).

Crystalline materials

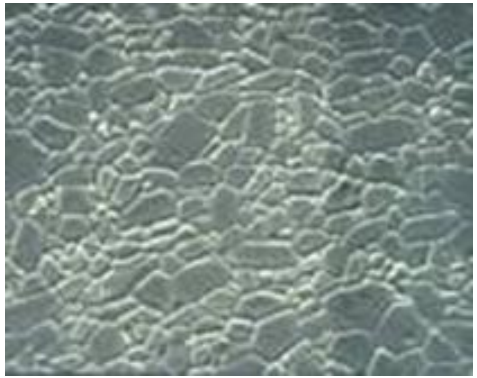




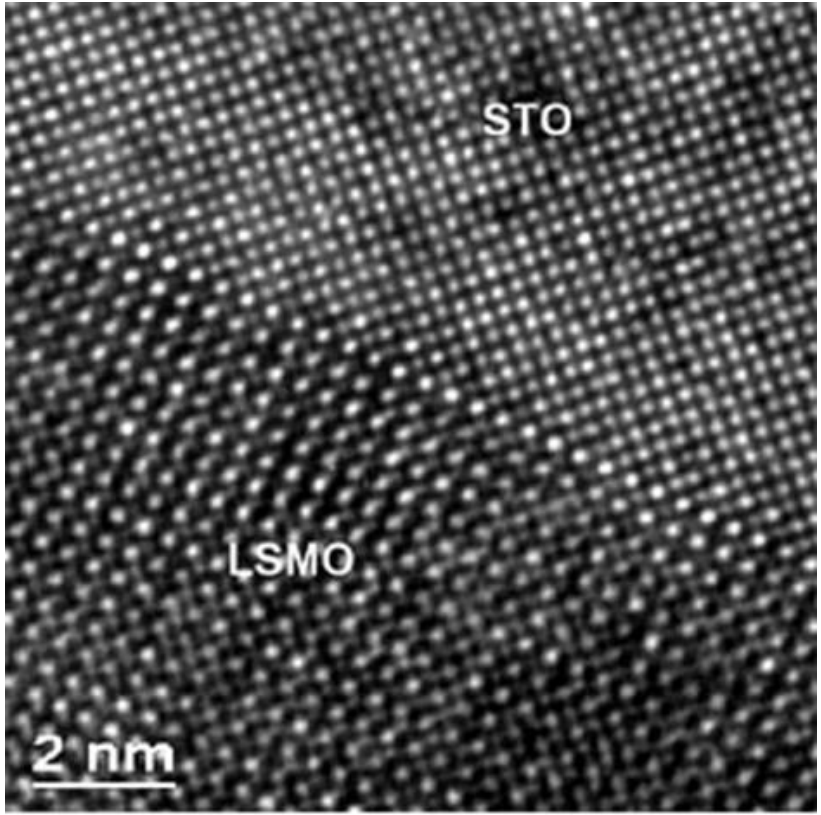
Single phase stainless steel



2-phase mild steel



Alumina



HRTEM of “single crystal” thin film Structure.

Crystallography

Study of this long range order.

- What are the types of arrangement possible?
- What kinds of order or symmetry elements are present?
- How do you explain all of these in a systematic fashion?

Note: The 230 types of order or space groups were mathematically derived much before it was proved that crystalline solids were ordered arrangements in 1891

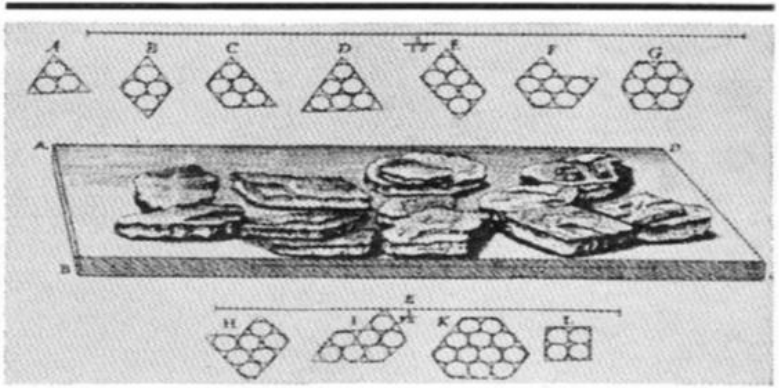
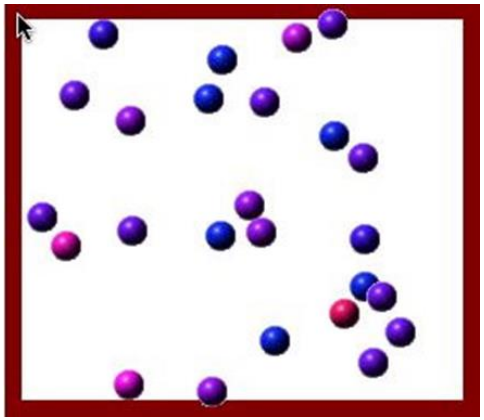


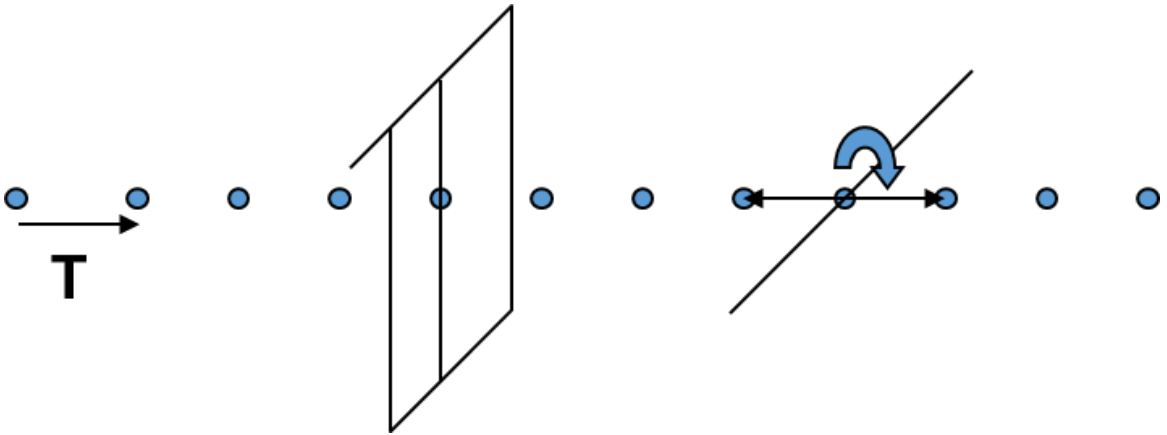
Fig. 3. The sphere-packing arrangements of Hooke (1665) to account for the natural shapes of crystals. (Courtesy of University of California Press).

What do you mean by order?

Disorder:

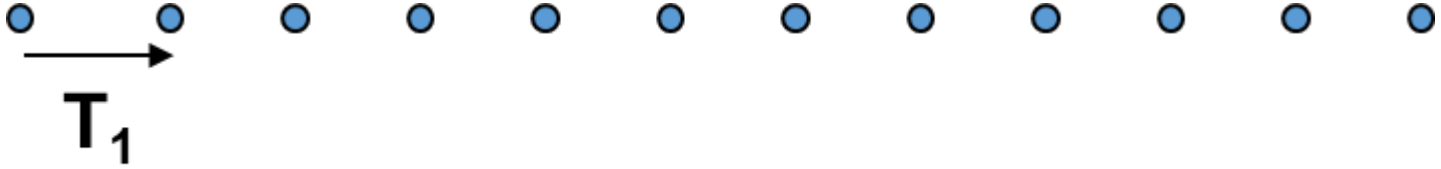


Order: Predictable



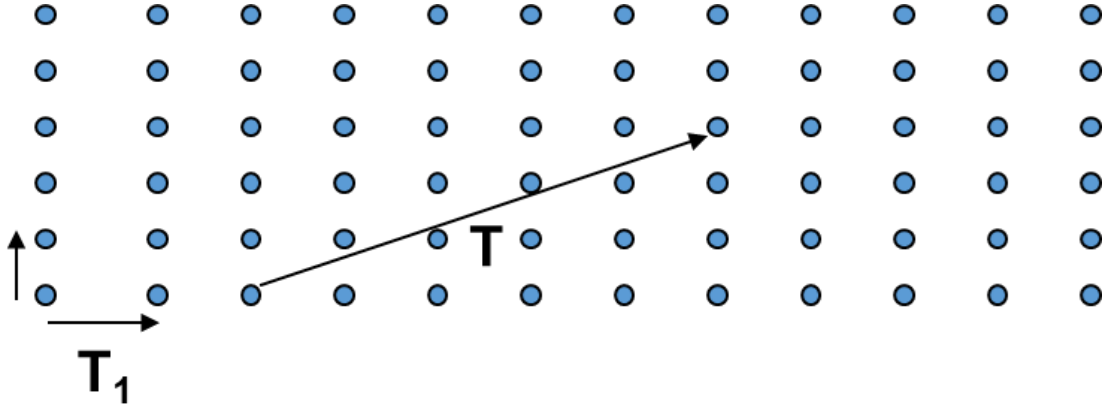
Ordered Arrays

1-D:



Ordered arrangements in 2-D can be obtained by translating the 1-D array in the second dimension by a vector T_2.

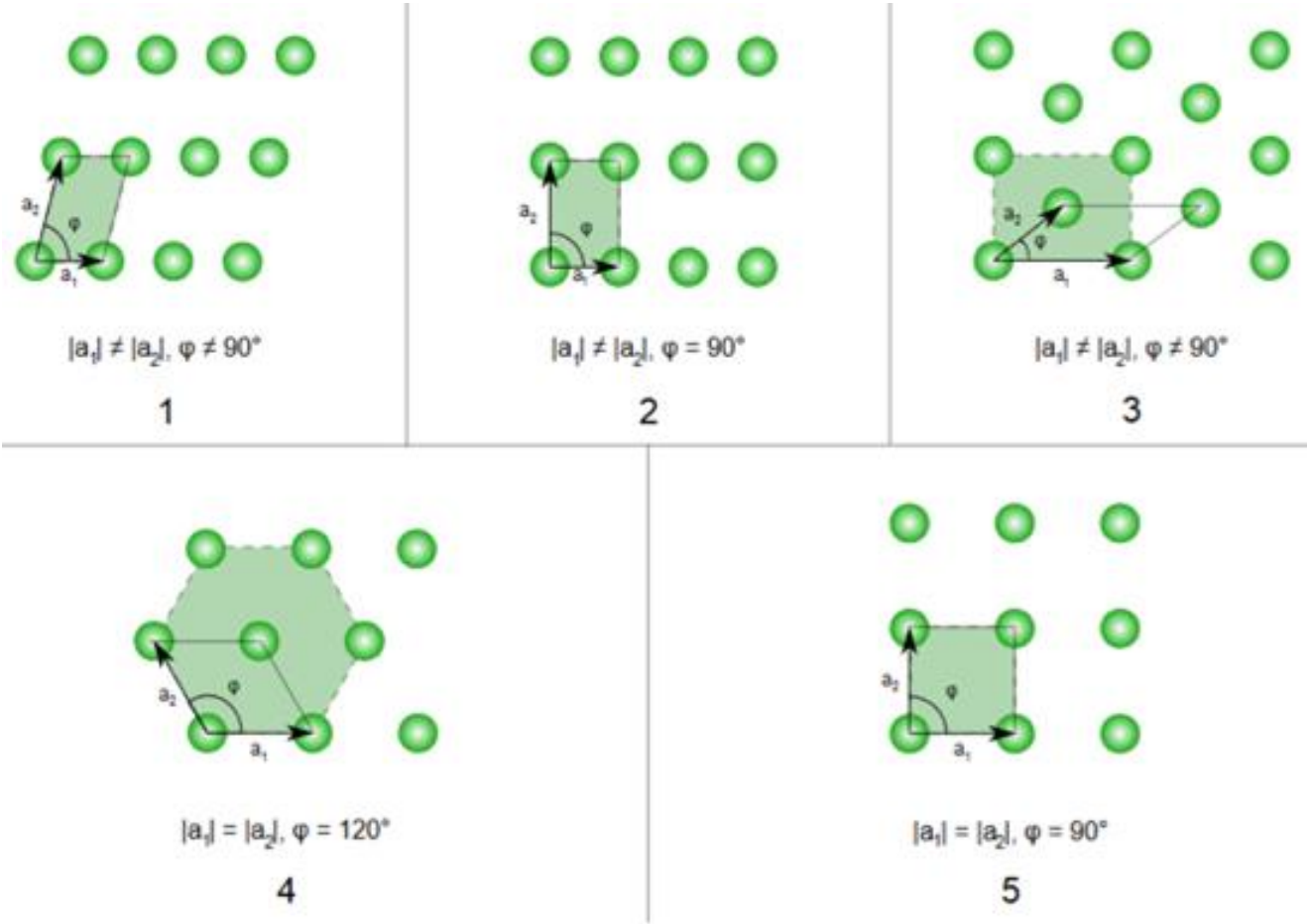
2-D:



$T = aT_1 + bT_2$; a and b are integers

Any symmetry element should be compatible with long range translational periodicity

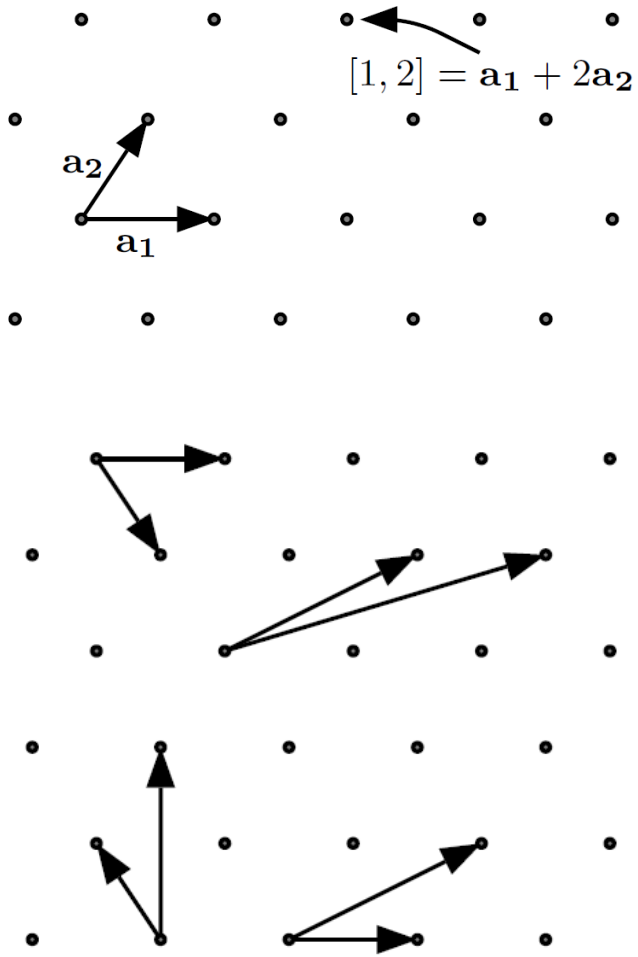
2-D Bravais Lattices



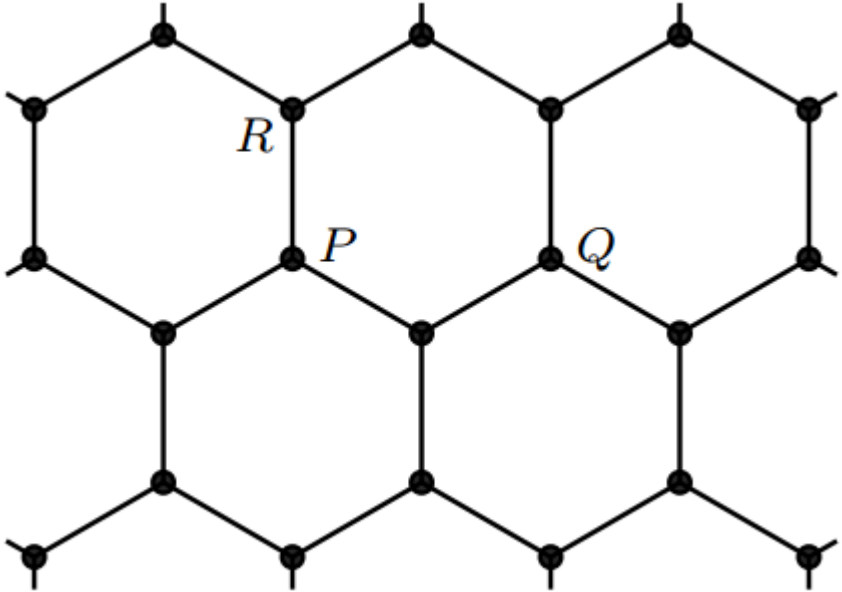
a ≠ b, γ ≠ 90	oblique
a ≠ b, γ = 90	rectangle
a ≠ b, γ = 90	Centered rectangular
a = b, γ = 120	Hexagon, Triangle
a = b, γ = 90	Square

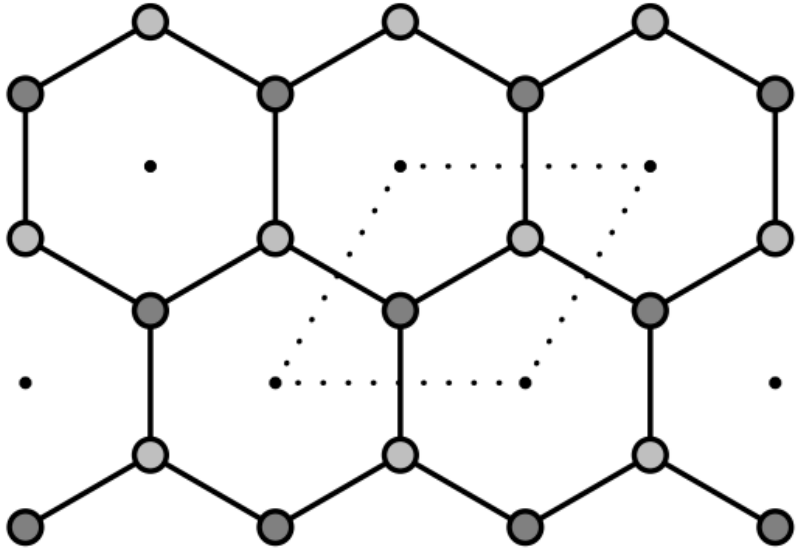
What is a Bravais Lattice?

- Lattice is an infinite set of points defined by integer sums of a set of linearly independent primitive lattice vectors (needn't be position of atoms)
- Lattice is a set of points where the environment of any given point is equivalent to the environment of any other given point
- Choice of primitive lattice vectors for a lattice is not unique



Vertices of Hexagonal comb (Can it be considered as a Bravais lattice?)

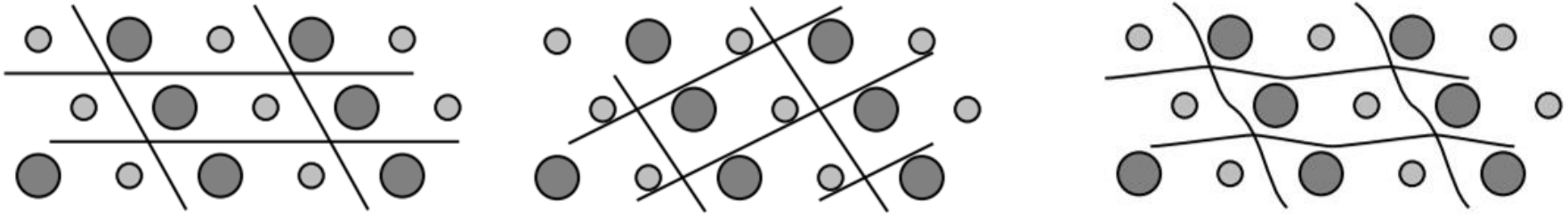




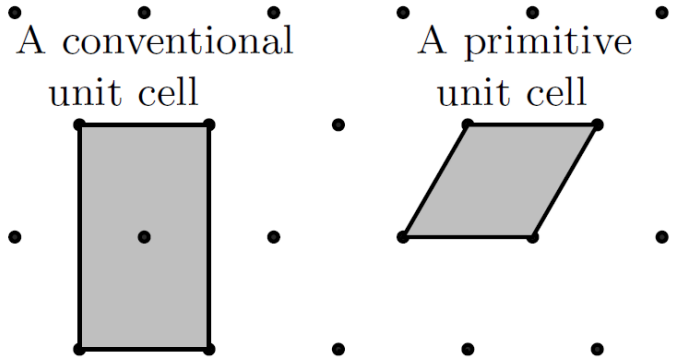
- Not all periodic arrangements of points are lattices
- For complicated arrangements of points, understand **unit cell**

Unit cell and primitive unit cell

- **Unit cell** is a region of space such that when many identical units are stacked together it tiles (completely fills) all of space and reconstructs the full structure.

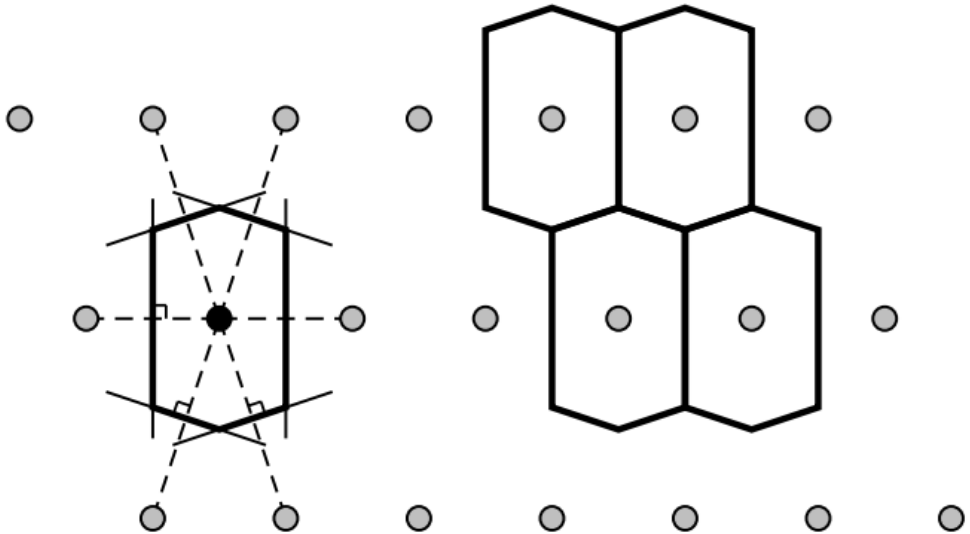


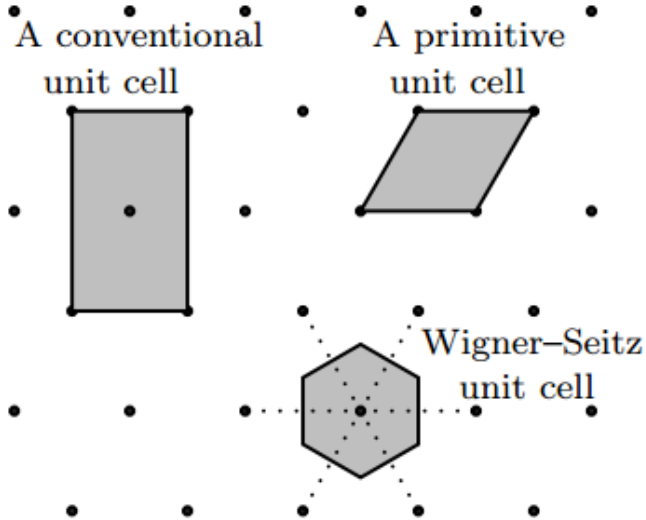
- **Primitive unit cell** for a periodic crystal is a unit cell containing exactly one lattice point.
- The choice of a unit cell or a primitive unit cell isn't unique



Wigner-Seitz primitive cell

Given a lattice point, the set of all points in space which are closer to that given lattice point than to any other lattice point constitute the **Wigner–Seitz cell** of the given lattice point.





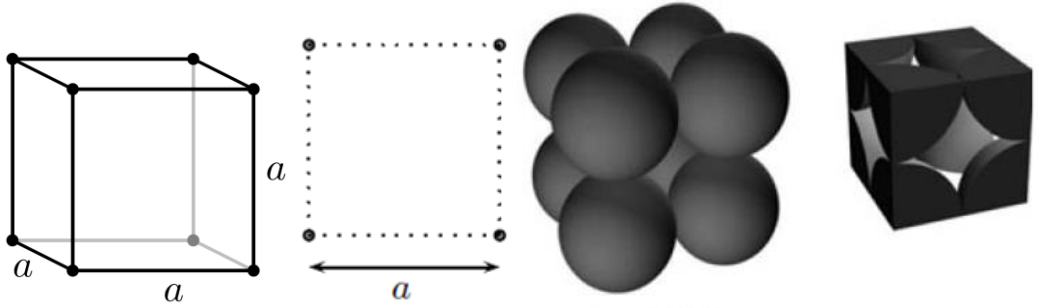
Volume or area of all primitive unit cells are the same (density of lattice points x volume of cell=1 for all such cells)

3-D Lattices: Cubic Lattices

Take 2-D square lattice & translate it by T_3 in third dimension

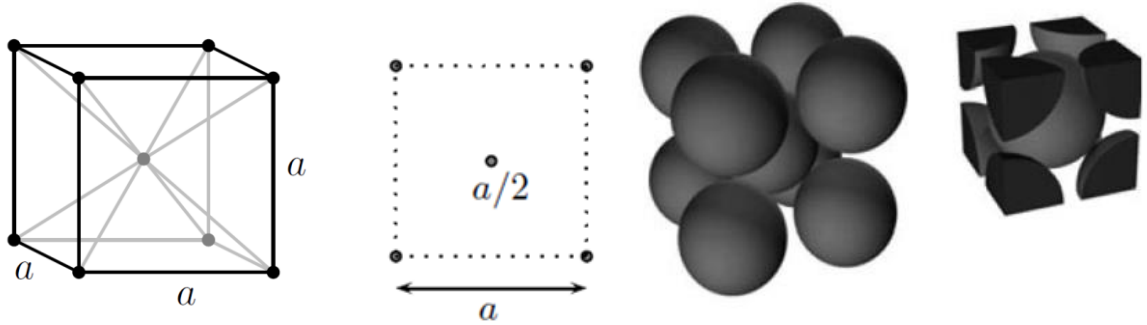
Cubic “P” or cubic-primitive lattice

Packing density= 50%



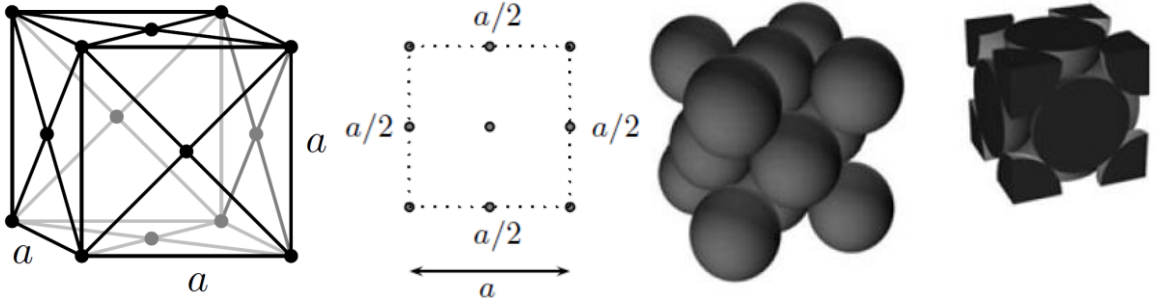
Body-Centered Cubic (bcc) Lattice

Packing density= 68% (Calculate)

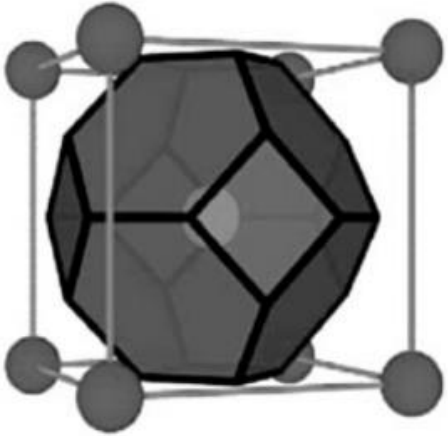


Face-Centered Cubic (fcc) Lattice

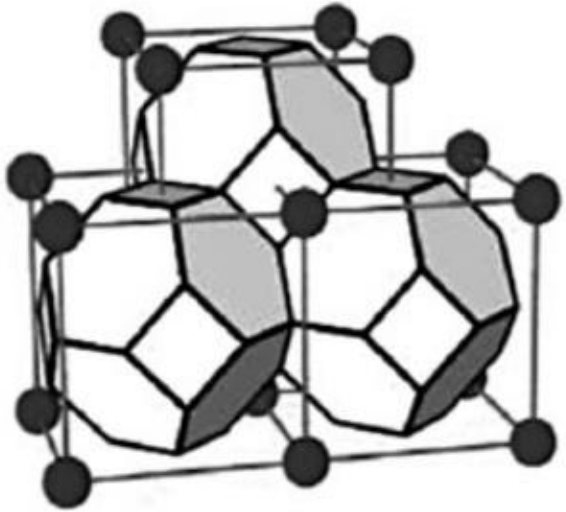
Packing density= 74% (Calculate)

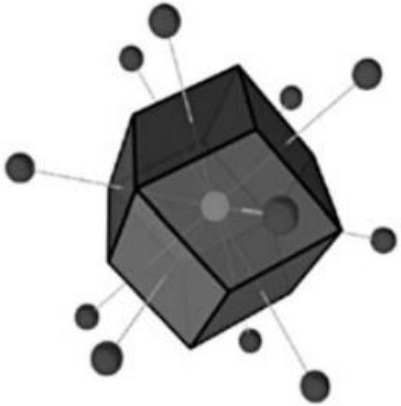


Wigner-Seitz unit cell

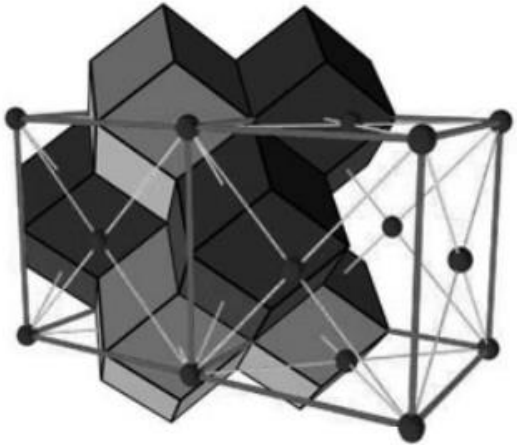


BCC Lattice
Truncated octahedron





FCC Lattice
Rhombic dodecahedron



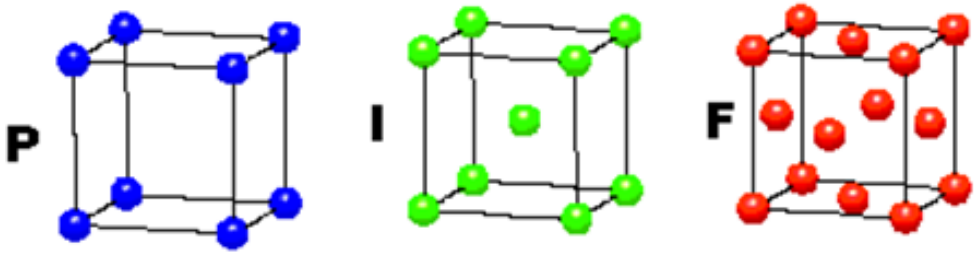
3D Bravais Lattices (1)

14 unique lattices, 7 crystal systems

CUBIC

$a = b = c$

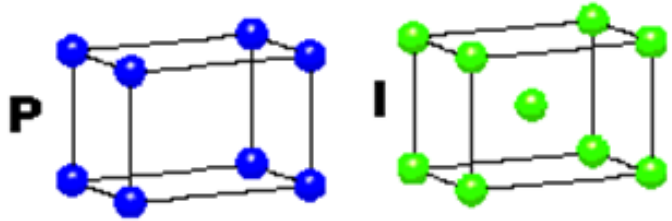
$\alpha = \beta = \gamma = 90°$



TETRAGONAL

$a = b \neq c$

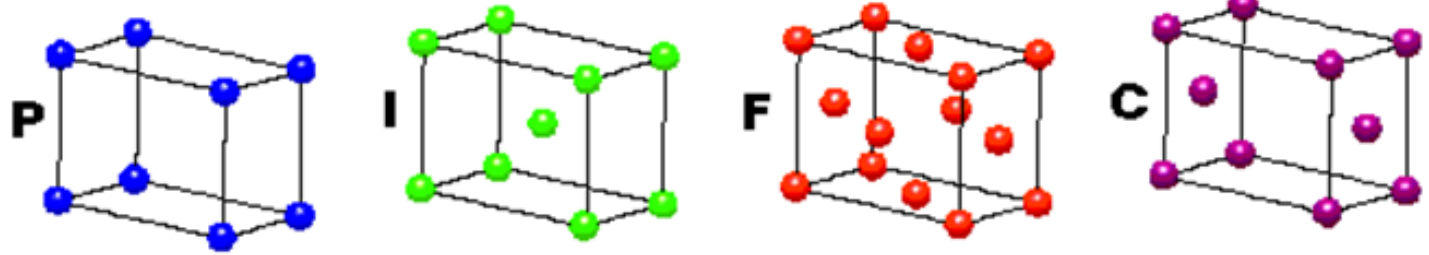
$\alpha = \beta = \gamma = 90°$



ORTHORHOMBIC

$a \neq b \neq c$

$\alpha = \beta = \gamma = 90°$



3D Bravais Lattices (2)

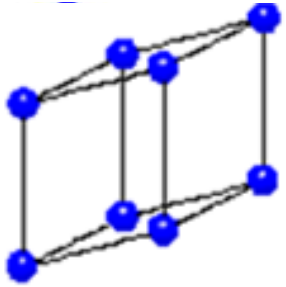
HEXAGONAL
$a = b \neq c$
$\alpha = \beta = 90°$
$\gamma = 120°$

P



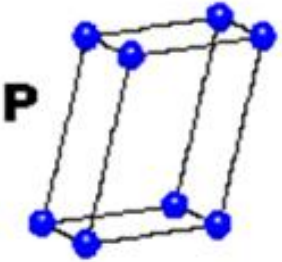
TRIGONAL
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$

P

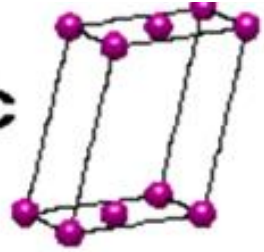


MONOCLINIC
$a \neq b \neq c$
$\alpha = \gamma = 90°$
$\beta \neq 120°$

P

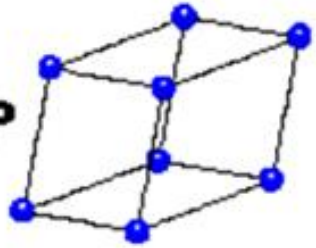


C



TRICLINIC
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$

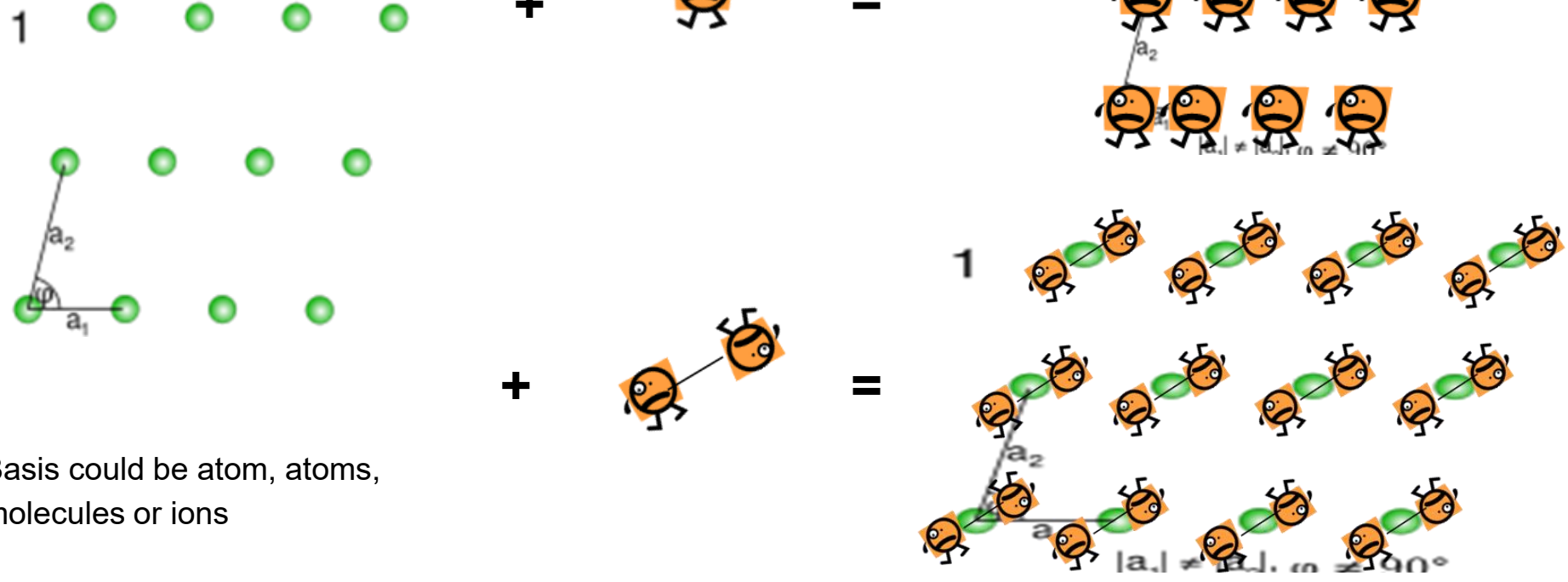
P



4 Types of Unit Cell
P = Primitive
I = Body-Centred
F = Face-Centred
C = Side-Centred
+
7 Crystal Classes
→ **14 Bravais Lattices**

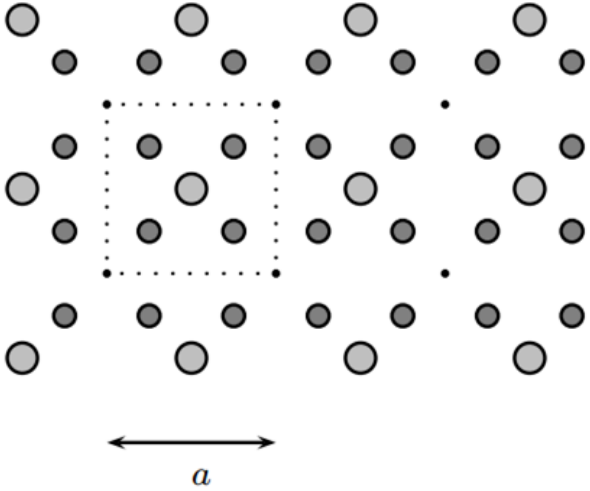
Lattice and Crystal Structure

Crystal Structure = Lattice + Motif (basis)



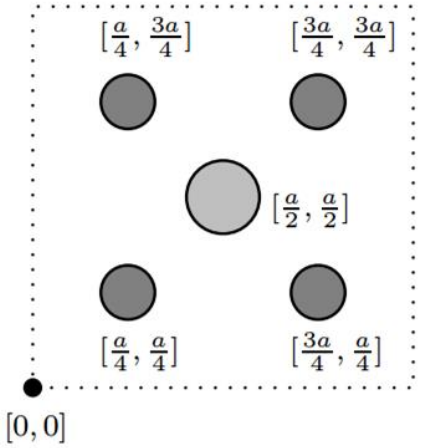
Basis could be atom, atoms, molecules or ions

Basis



The description of objects in the unit cell with respect to the reference lattice point in the unit cell is known as a basis

Reference points (the small black dots in the figure) forming the square lattice have positions $\mathbf{R}_{[n_1\,n_2]} = [a\,n_1, a\,n_2] = a\,n_1\hat{\boldsymbol{x}} + a\,n_2\hat{\boldsymbol{y}}$



$$\mathbf{R}^{light-gray}_{[n_1\,n_2]} = [a\,n_1, a\,n_2] + [a/2, a/2]$$

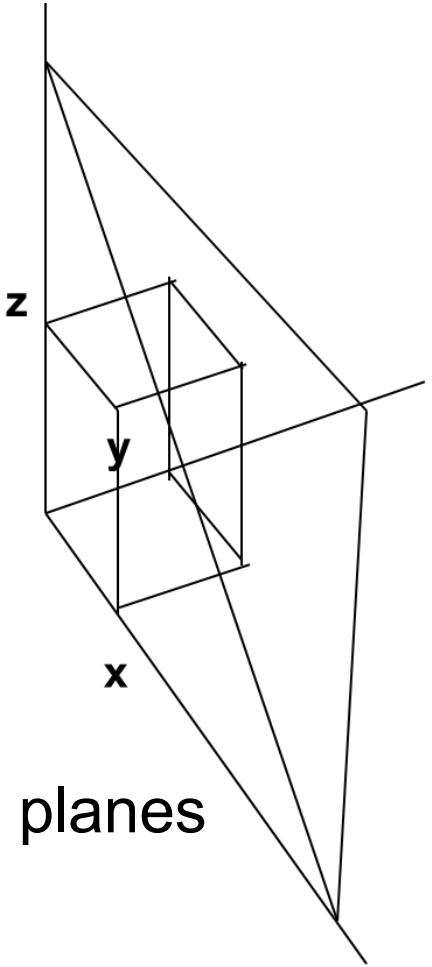
$$\begin{aligned}
\mathbf{R}^{dark-gray1}_{[n_1\,n_2]} &= [a\,n_1, a\,n_2] + [a/4, a/4] \\
\mathbf{R}^{dark-gray2}_{[n_1\,n_2]} &= [a\,n_1, a\,n_2] + [a/4, 3a/4] \\
\mathbf{R}^{dark-gray3}_{[n_1\,n_2]} &= [a\,n_1, a\,n_2] + [3a/4, a/4] \\
\mathbf{R}^{dark-gray4}_{[n_1\,n_2]} &= [a\,n_1, a\,n_2] + [3a/4, 3a/4]
\end{aligned}$$

Need for indexing Crystallographic Planes and Directions

- Only few properties (eg: density) are truly isotropic
- Some Properties that are not isotropic:
  - Mechanical properties (elastic constants, cleaving)
  - Electrical properties (electron transport, mobility, defect states)
  - Chemical reaction rates

Lattice Plane and Miller Indices

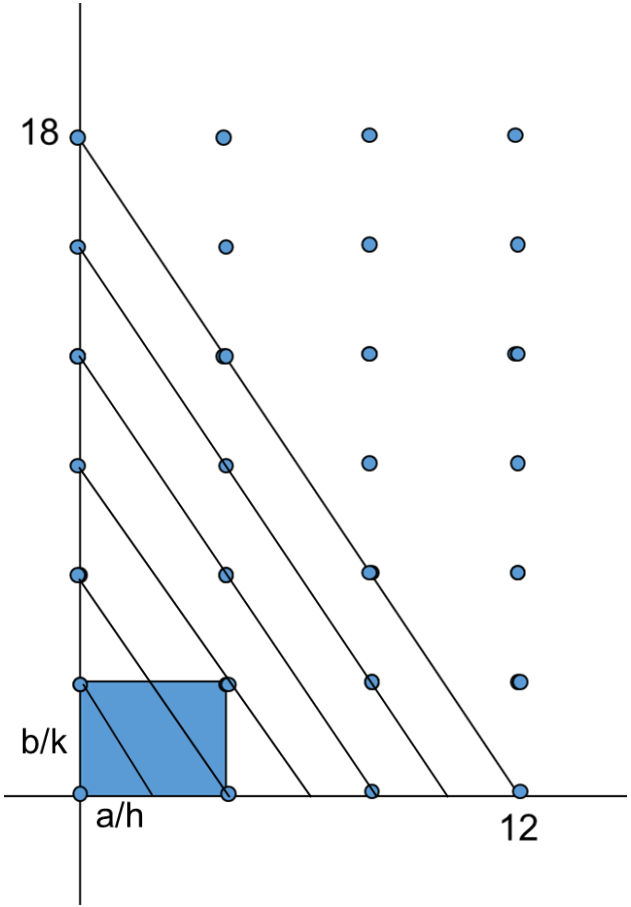
- How to find Miller Indices (h, k, l) for a plane?
  - ❖ Origin shouldn't be on the plane
  - ❖ Determine intercepts x, y, z (10, 12, 18)
  - ❖ Divide by unit cell parameters a, b, c (2, 4, 3)
    x/a, y/b, z/c (5, 3, 6)
  - ❖ Take the inverse (to get rid of ∞)
    a/x, b/y, c/z (1/5, 1/3, 1/6)
  - ❖ Find smallest set of integers
    h, k, l such that h:k:l::a/x:b/y:c/z (6, 10, 5)



- ❑ All parallel planes have the same (hkl) indices
- ❑ {hkl} to denote crystallographic equivalent (hkl) planes

$$\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$$

Miller Indices example



- Plane intercepts x, y (12, 18)
- Unit cell parameters a, b (4, 3)
  x/a, y/b (3, 6)
- Take the inverse
  a/x, b/y (1/3, 1/6)
- h:k::a/x:b/y (2, 1)

❖ (h, k) = (2, 1)

❖ a/h, b/k = 2, 3 (Intercepts of plane closest to origin)

Miller Indices: Identify the planes

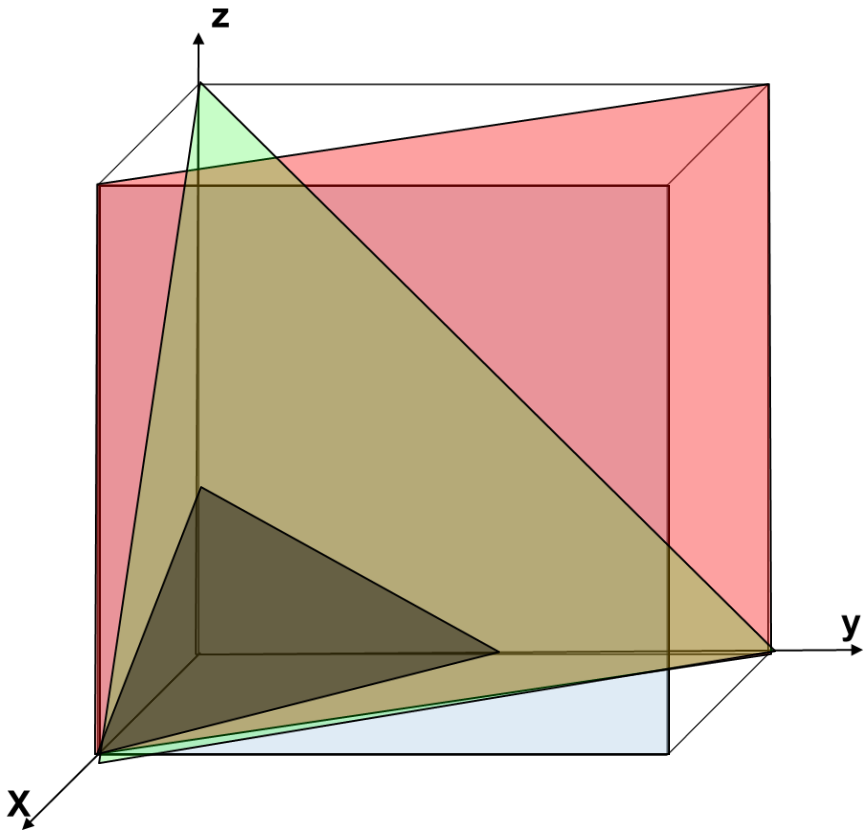
(100)

(110)

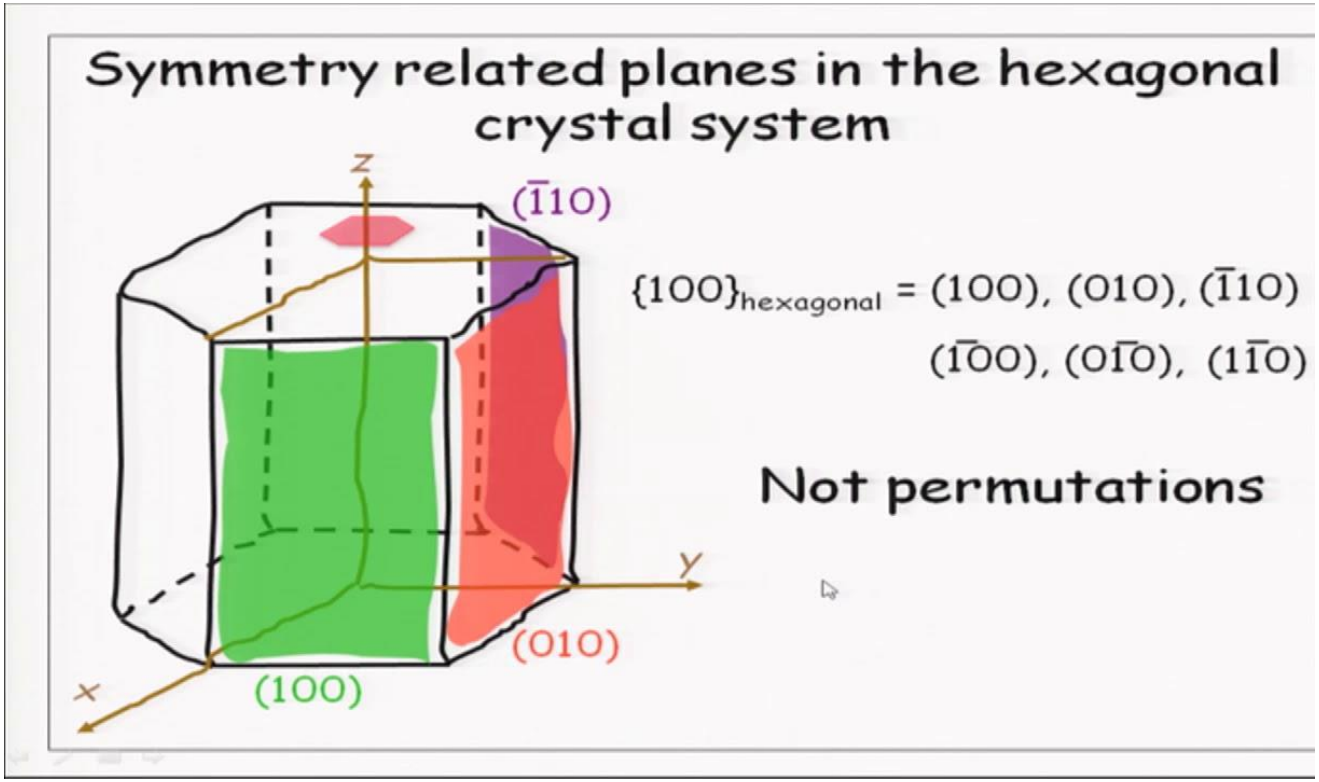
(111)

(200)

(123)



Miller Indices for Hexagonal Systems



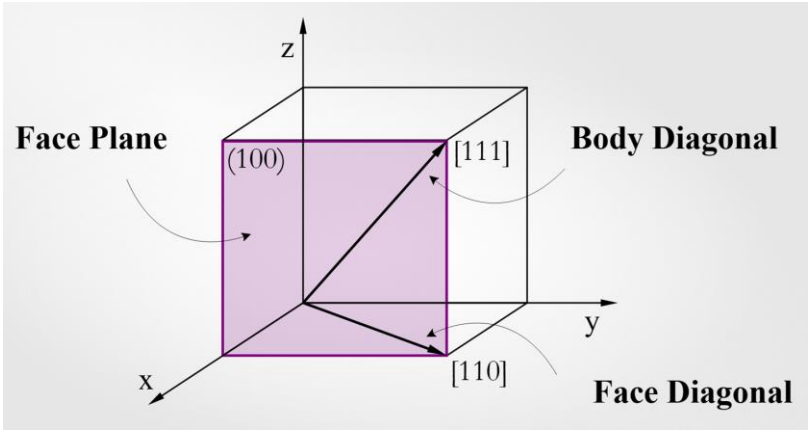
Please note here the x & y axis aren't 90 deg apart. They are 120 deg apart to follow the six-fold symmetry of the crystal. The intercepts and Miller indices given are for this coordinate axis.

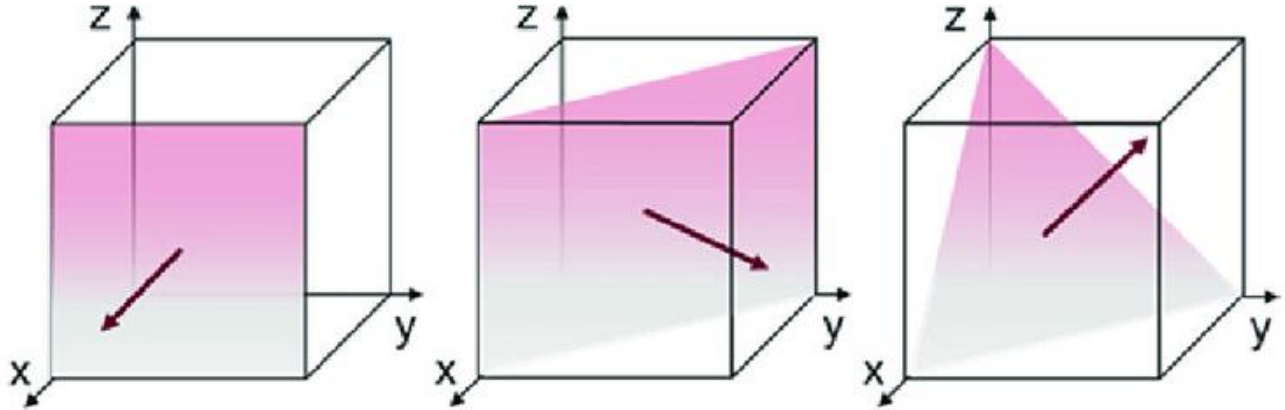
Here the Miller indices aren't permutations like the sides of a cube using cubic system

This issue is resolved by using Miller-Bravais system(hkil) (not included in this course)

Directions [h,k,l]

- Determined by coordinates of point through which the direction vector passes and represented by the smallest set of integers.
- In cubic crystals, [hkl] is perpendicular to the (hkl) plane





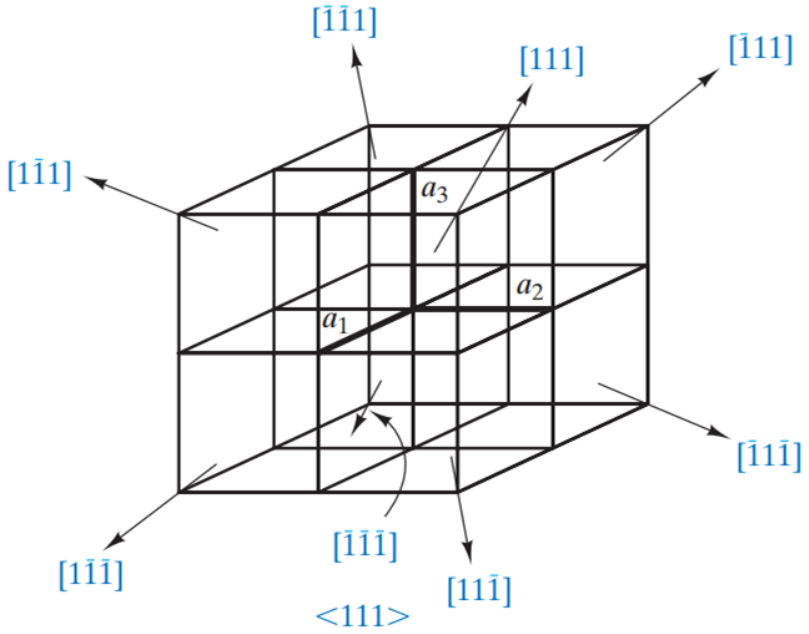
Directions [h,k,l]

- <hkl> to denote crystallographic equivalent [hkl] directions

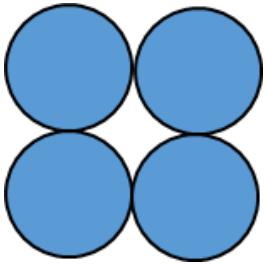
$$\langle 111 \rangle = [\bar{1}11], [1\bar{1}1], [11\bar{1}], [\bar{1}\bar{1}1], [1\bar{1}\bar{1}], [\bar{1}1\bar{1}], [\bar{1}\bar{1}\bar{1}]$$

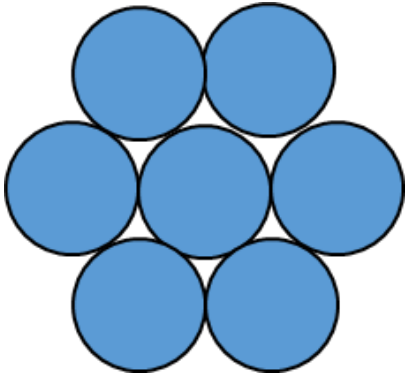
- A bar on top is used to denote negative direction or intercepts
- Put a bar on top of all to reverse the direction
- Angle between two directions can be found by

$$\cos(\theta) = \frac{h_1 h_2 + k_1 k_2 + l_1 l_2}{[(h_1^2 + k_1^2 + l_1^2)(h_2^2 + k_2^2 + l_2^2)]^{1/2}}$$



Atomic packing in solids

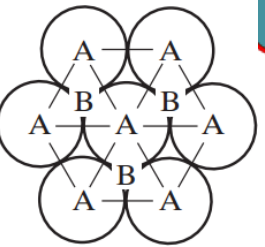


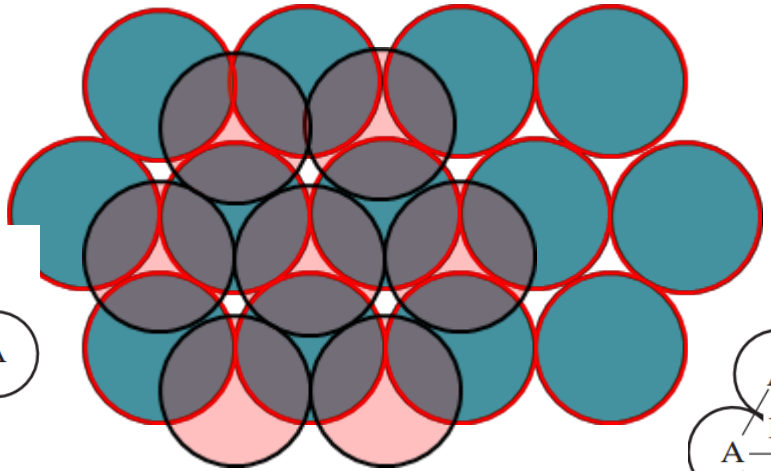


2D

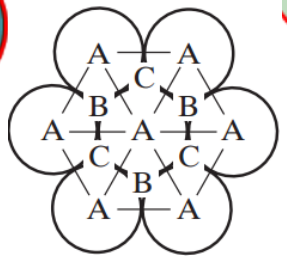
There are 2 things in the drawing: Atom and voids

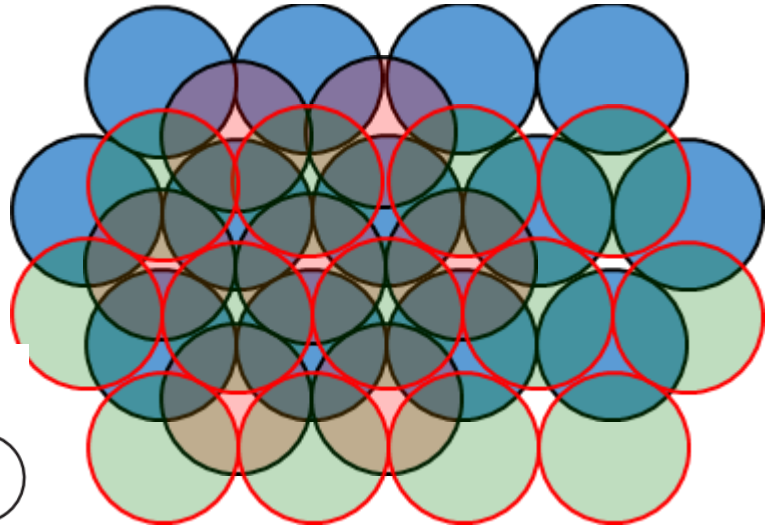
3D





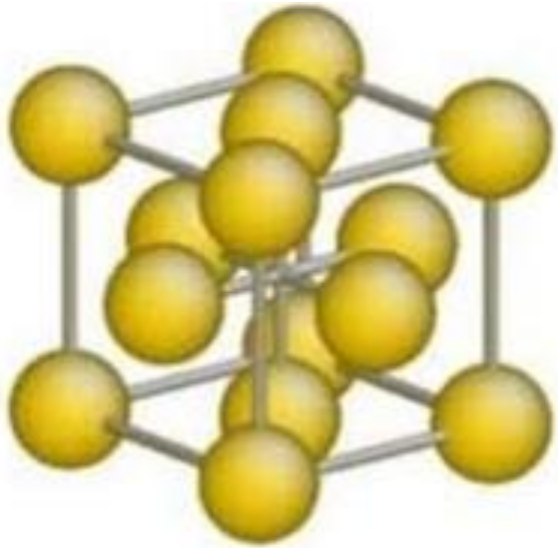
ABAB.. → Hexagonal close packed



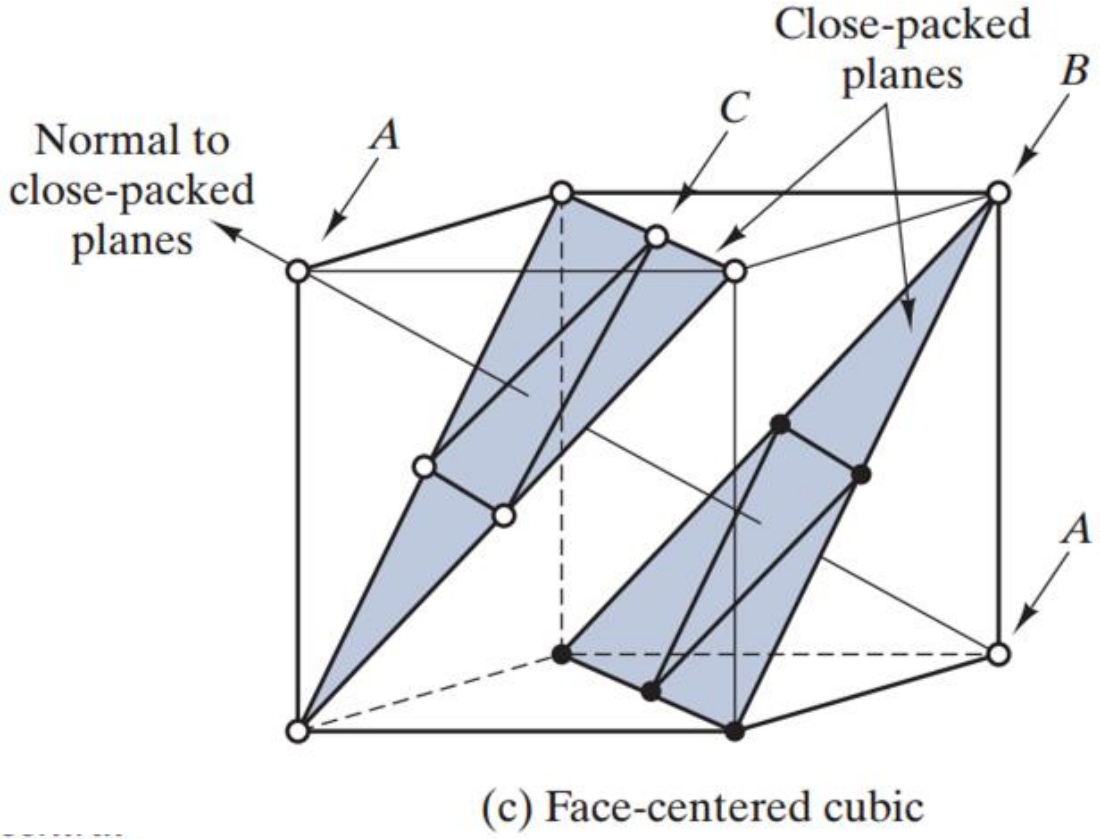


ABCABC.. → FCC

FCC Stacking

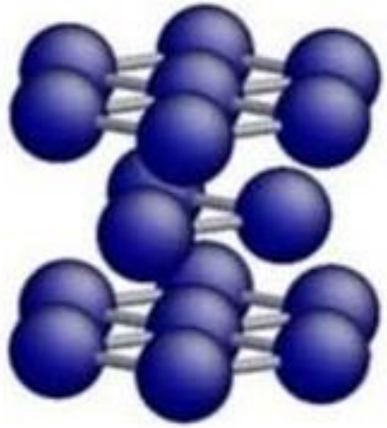


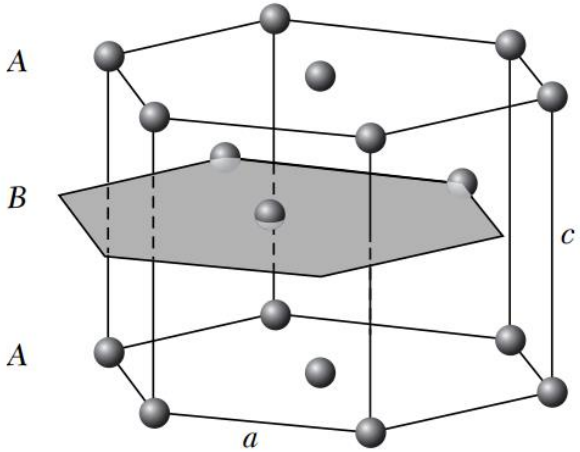
Stacking is ABCABC along [111] direction of FCC and not along [100]

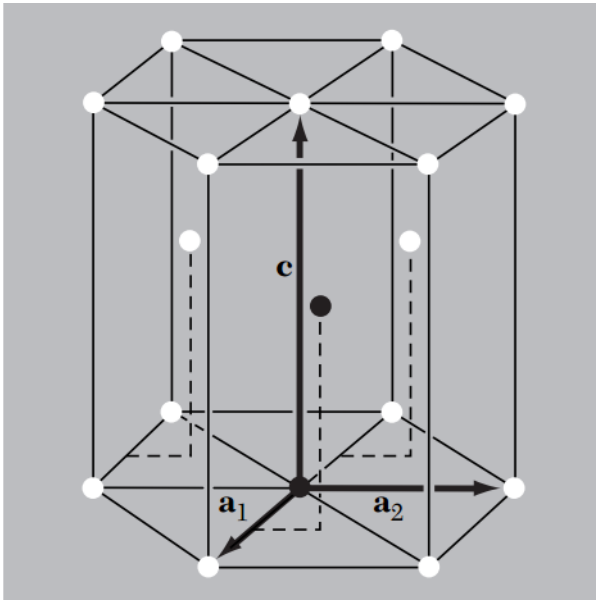


(c) Face-centered cubic

HCP Stacking



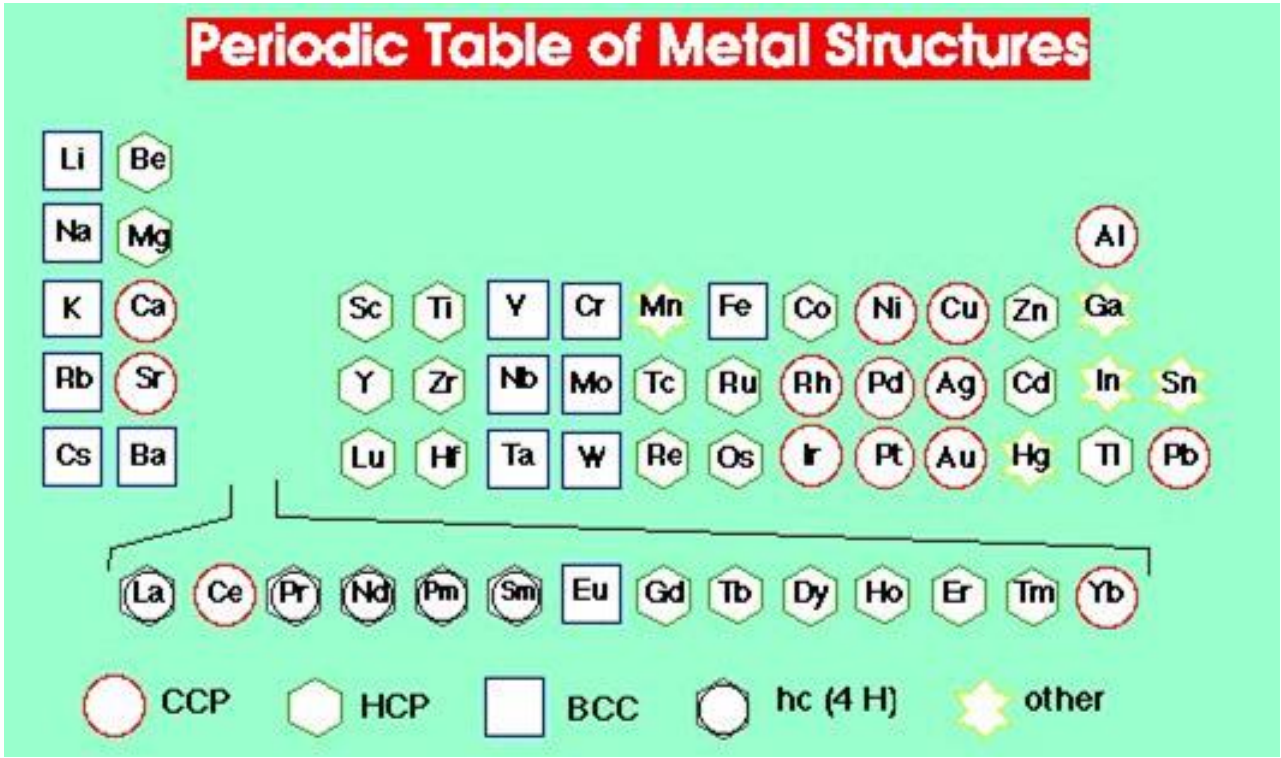




The atom positions in HCP structure do not constitute a space lattice. The space lattice is simple hexagonal with a basis of two identical atoms associated with each lattice point.

Figure 21 The primitive cell has $a_1 = a_2$, with an included angle of 120°. The c axis (or $\mathbf{a}_3$) is normal to the plane of $\mathbf{a}_1$ and $\mathbf{a}_2$. The ideal hcp structure has $c = 1.633\,a$. The two atoms of one basis are shown as solid circles. One atom of the basis is at the origin; the other atom is at $\frac{2}{3}\,\frac{1}{3}\,\frac{1}{2}$, which means at the position $\mathbf{r} = \frac{2}{3}\mathbf{a}_1 + \frac{1}{3}\mathbf{a}_2 + \frac{1}{2}\mathbf{a}_3$.

Structure of Metals



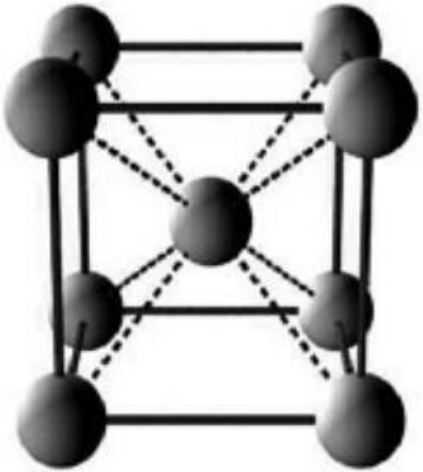
55/94 (natural) elements in the periodic table have BCC, FCC, HCP

Some real examples

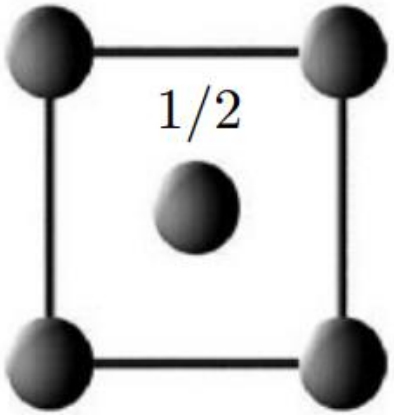
Sodium (Na)

Lattice = Cubic-I (bcc)

Basis = Na at [000]



Plan view
unlabeled points at $z = 0, 1$

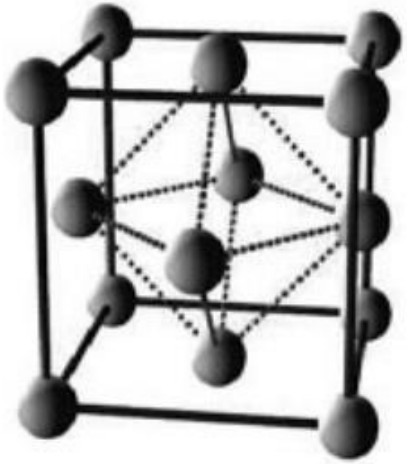


Some real examples

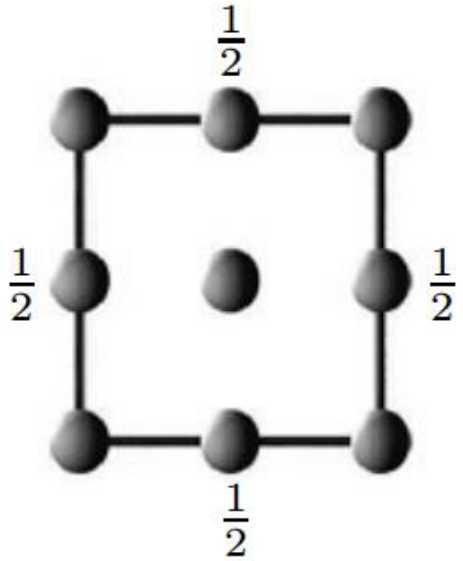
Copper(Cu)

Lattice = Cubic-F (fcc)

Basis = Cu at [000]



Plan view
unlabeled points at $z = 0, 1$



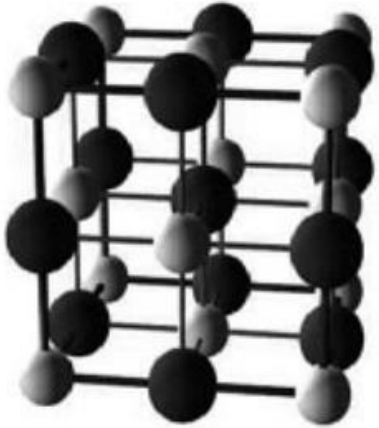
Some real examples

Sodium Chloride (NaCl)

Lattice = Cubic-F (fcc)

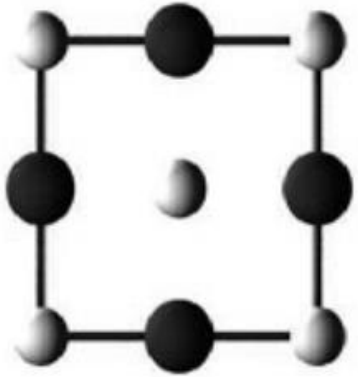
Basis = Na at [000]

and Cl at $[\frac{1}{2}\frac{1}{2}\frac{1}{2}]$

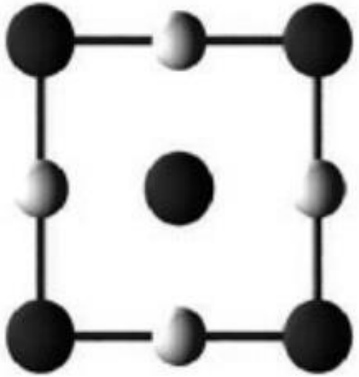


Plan view

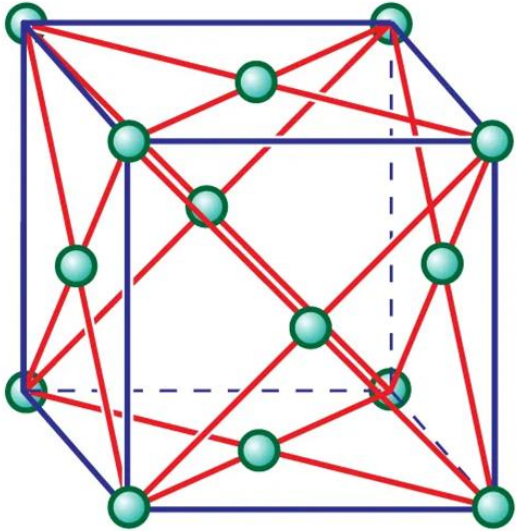
$z = 0, 1$ layer



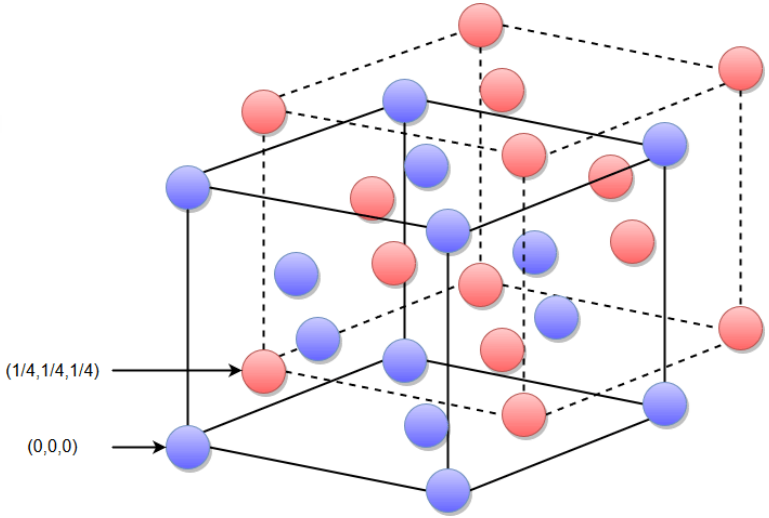
$z = \frac{1}{2}$ layer



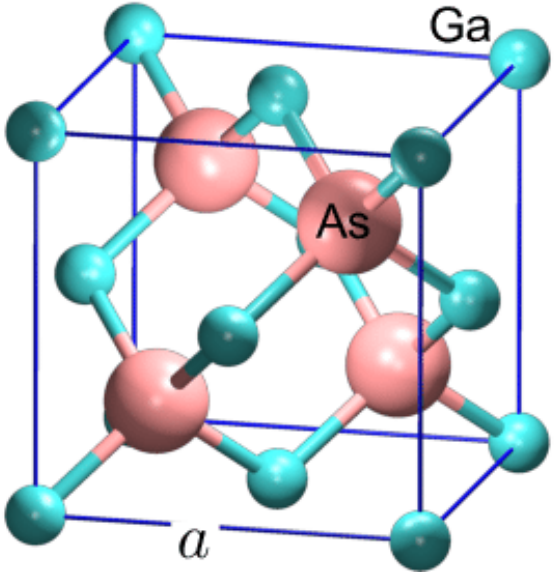
Some real examples



FCC lattice
e.g. Al, Cu

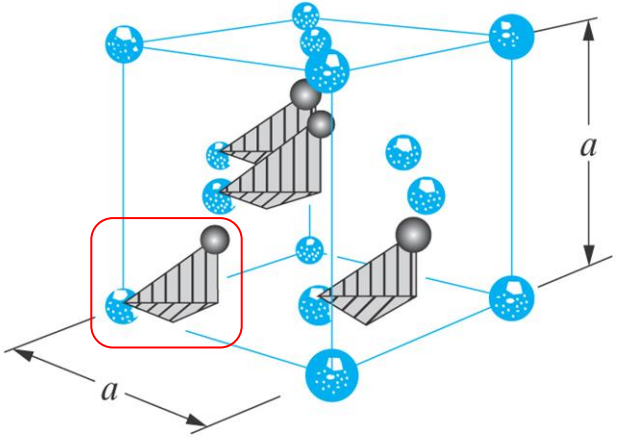


Diamond lattice (Si, Ge)
Two face-centered cubic (FCC) lattices inter-penetrating with another. The 2nd FCC lattice is shifted by

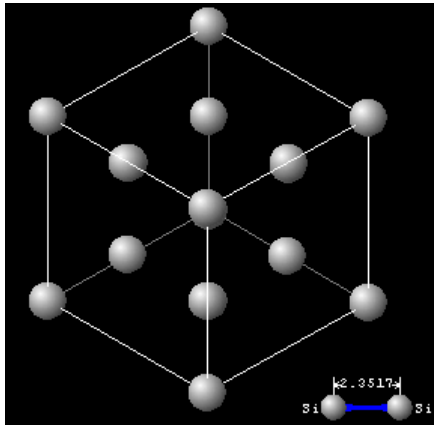


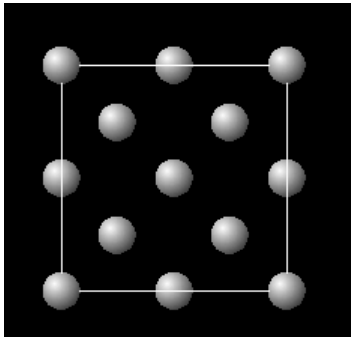
Zinc Blende lattice (GaAs, SiC, GaN, InP)
Diamond lattice with alternating atoms in each FCC

Si crystal viewed in different directions

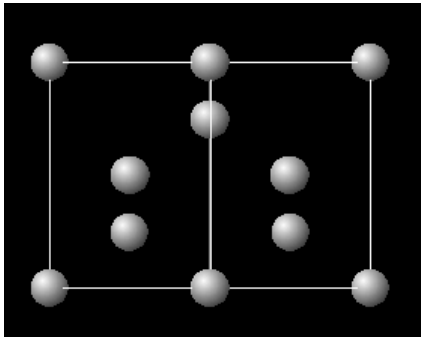


FCC with a 2 point basis
Primitive cell contains
2 atoms





<100>



<110>

<111>

If a=5.43Å, find the density of atoms in Silicon lattice